DEPLOYMENT REPORT

Project Title

Blockchain-Based Academic Certificate Verification System

Task 2: Deployment

Application Technology: Flask (Python)

Application Type: Web Application

1. Objective

The objective of this task is to deploy the Flask-based academic certificate verification application and verify that the application works correctly in a deployed environment.

2. Project Overview

The project is a Blockchain-Based Academic Certificate Verification System. The application allows users to register and log in, verify academic certificates manually using student details and Certificate ID, upload PDF certificates for automated verification, and use an AI Assistant for application guidance.

3. Deployment Preparation

The Flask application was prepared for deployment by organizing the Python application code, HTML templates, student data file, and required dependencies. The application uses Flask as the backend framework and includes PyPDF2, Pandas, pdf2image, Tesseract OCR and OpenAI API functionality.

4. Application Deployment

The Flask application was deployed to a cloud hosting environment. After deployment, the application was accessed using the generated application URL. The deployed application was checked to confirm that the backend and frontend components were working correctly.

5. Deployment Verification Activities

Test No.	Functionality	Verification Activity	Result
1	Application Access	Open deployed application URL	PASS
2	Registration	Create a new user account	PASS
3	Login	Login using valid credentials	PASS
4	Dashboard	Access the verification dashboard	PASS
5	Manual Verification	Enter student name and Certificate ID	PASS
6	PDF Upload	Upload a certificate PDF	PASS
7	Certificate Processing	Verify text/OCR processing	PASS
8	AI Assistant	Send a question to AI Assistant	PASS
9	Logout	Logout from the application	PASS

6. Deployment Verification

The deployed application was verified by accessing the live application and checking the main application features. The registration and login modules were checked successfully. The dashboard was accessed after login. Manual certificate verification and PDF certificate processing were verified. The AI Assistant was also tested to confirm that the chat functionality was working.

7. Deployment Result

The application was successfully deployed and the deployed application functionality was verified. The main application modules responded as expected in the deployed environment.

8. Conclusion

The Blockchain-Based Academic Certificate Verification System was successfully deployed. The application was accessed in the deployed environment and its major features were verified. Registration, login, certificate verification, PDF processing and AI Assistant functionality were tested successfully. Therefore, the deployment task was completed successfully.